

Ain e Muhammad
Senior Software Engineer | Full-Stack | AI & 3D Systems
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EXPERTISE SUMMARY

- Senior Software Engineer with extensive experience in Full-Stack development (React.js / Python), Serverless Architecture (AWS), and real-time 3D environments.
- Currently pursuing an M.S. in Computer Science (AI), specializing in Machine Learning, Computer Vision, and Generative 3D workflows.
- Hands-on experience architecting Generative AI pipelines, including leveraging state-of-the-art diffusion and 3D reconstruction models (Meshy, Tripo 3D, Hunyuan3D) alongside complex object/camera placement algorithms.
- Delivered production-ready full-stack applications with React.js, Python, and serverless AWS architectures, supporting high-concurrency workloads with minimal latency.
- Specialized in 3D Systems & Graphics Development, utilizing Unreal Engine for cinematic environments and Three.js / React Three Fiber / Blender for web-based interactive product configurators.
- Expertise in Cloud Infrastructure, specifically designing and deploying high-concurrency AWS Lambda functions, API Gateway, S3, and RDS/PostgreSQL databases.
- Proven record in Data Engineering, including dataset curation, schema design, SQL optimization, and building automated validation pipelines.
- Led and mentored junior engineers, improving team productivity and code quality through structured code reviews, CI/CD implementation, and unit testing.

EDUCATION

State University of New York (SUNY) - Albany, NY Masters in Computer Science (AI) | Expected May 2027

- **Relevant Coursework:** Machine Learning, Computer Vision, Advanced Algorithms and Data Structures.

National University of Sciences and Technology (NUST), Bachelor of Science in Software Engineering | July 2021

- **Relevant Coursework:** Artificial Intelligence, Data Warehousing and Data Mining, Data Structures and Algorithms, Database Systems, Digital Image Processing, Software Engineering, Probability and Statistics, Object-Oriented Programming, Data Structures & Algorithms, Web Engineering, Software Architecture.

WORK EXPERIENCE

3D Creative Engineering Intern – Sony Immersive Music Studio | June 2026 – Aug 2026

- **Generative 3D Pipelines:** Architected an end-to-end generative pipeline that reconstructs immersive 3D scenes from 2D images by orchestrating SOTA image segmentation (masks), diffusion-based image generation, and 2D-to-3D asset generation models.
- **Spatial & Camera Automation:** Programmed automated camera placement and layout algorithms utilizing cutting-edge spatial computing and object placement research papers, ensuring mathematically precise asset distribution within a 3D space.
- **Real-time Graphics & Environments:** Engineered interactive 3D environments and cinematic sequences within Unreal Engine, managing real-time rendering, material shaders, and complex environmental lighting.
- **AI-Assisted Scene Generation:** Designed procedural environment generation workflows leveraging Florence-2 object detection, depth-map reconstruction, Blender Python automation, and AI-generated assets to transform concept art into structured 3D scenes.
- **Virtual Production Systems:** Developed Blender-to-Unreal workflows integrating AI-generated assets, automated scene assembly, lighting analysis, camera orchestration, and cinematic rendering pipelines for immersive content creation.

- **AI Tool Development:** Built proof-of-concept Unreal Engine tooling enabling concept-art-to-3D asset generation directly within the editor, supporting local AI inference and artist-controlled asset creation workflows.
- **Scene Optimization & Rendering:** Optimized large-scale 3D environments containing hundreds of assets through instancing strategies, mesh audits, lighting refinement, and rendering pipeline improvements for real-time performance.
- **AI Agent Workflows:** Developed custom AI-assisted tooling and workflows using Claude Code, MCP integrations directly into blender, Blender automation, and Unreal Engine pipelines to accelerate environment creation, lighting design, cinematography, and content iteration.

RAG Evaluation & Benchmark Engineer – HADI (ChatLLM.ai) | Jun 2026 – Aug 2026

- **RAG Evaluation Framework:** Designed and built an evaluation framework for a production RAG-based chatbot, defining measurable criteria across retrieval accuracy, citation correctness, and hallucination resistance in collaboration with the engineering team.
- **Data Pipeline Engineering:** Built automated data pipelines to pull and structure ~6,200 source records and ~36,500 reference records from external APIs and unstructured PDF sources, using column-aware and font-aware parsing to handle complex multi-column document layouts.
- **Automated Test Harness:** Developed a Python-based scoring harness to run benchmark suites against a live LLM API, with configurable automated checks for required-fact coverage, forbidden-claim detection, and citation presence.
- **Cross-functional Collaboration:** Partnered directly with the CTO and technical stakeholders to scope evaluation methodology, resolving ambiguity in system requirements through iterative, structured technical discussions.

Graduate Grader - State University of New York - Albany, NY | Jan 2026 – May 2026

- **Academic Support:** Grade assignments and exams for Discrete Structures, providing feedback on mathematical reasoning and problem-solving.
- **Course Coordination:** Support instructor with administration and provide insights into student challenges to improve curriculum delivery.

ALL3D - San Francisco, USA (Remote), May 2021 – Aug 2025

Software Engineer → Senior Software Engineer → Lead Software Engineer

Lead Software Engineer | Apr 2024 – Aug 2025

- **Client-Facing Technical Leadership:** Led delivery of 3D configurators for enterprise clients (**Walmart, Rough Country, Telescope Casual Furniture**), owning requirements, demos, and end-to-end technical execution, reducing delivery iteration cycles by ~20%.
- **Front-End Architecture Standardization:** Directed front-end architecture across multiple enterprise configurator products, standardizing CRUD-based data flows for product catalogs, user configurations, and saved sessions to ensure consistency across client deployments.
- **Codebase Standardization & Maintainability:** Introduced coding standards, reusable patterns, and documentation practices that improved onboarding speed and reduced long-term maintenance overhead.
- **Stakeholder Communication & Technical Demos:** Delivered technical walkthroughs and demos to non-technical stakeholders, improving transparency and accelerating decision-making during project cycles.
- **Team Leadership & Execution:** Managed a cross-functional team of engineers and 3D artists, improving delivery throughput through structured task ownership and code review processes.
- **3D Rendering Optimization:** Built React Three Fiber systems leveraging GPU acceleration, GLTF compression, and LOD techniques, improving render performance and reducing load times.
- **Engineering Mentorship:** Mentored engineers, enforced architectural standards, and led code reviews, reducing production issues and improving system reliability.
- **Reusable Component Systems:** Designed reusable front-end component libraries, including form inputs, modals, and config panels, with built-in create/read/update/delete patterns, reducing feature development time across client-specific configurators.

Senior Software Engineer | Feb 2023 – Apr 2024

- **Serverless Architecture:** Architected AWS Lambda and API Gateway systems with RDS (PostgreSQL), improving scalability and reducing backend latency.
- **Frontend Scalability:** Developed React systems with asset streaming and lazy loading, improving initial load performance by ~35% under high concurrency.
- **3D Rendering Pipelines:** Owned backend systems for dynamic rendering (camera, lighting, scene configs), enabling flexible and reusable visualization workflows.
- **AI Experimentation:** Collaborated with AI teams on early-stage generative pipelines, improving model output quality through iterative evaluation and feedback loops.
- **System Reliability Engineering:** Implemented structured validation and error handling, reducing runtime failures and improving API stability.
- **Admin Tooling & CRUD Interfaces:** Built and maintained full CRUD interfaces for internal admin tools, enabling non-technical staff to create, edit, and manage 3D asset metadata, product configurations, and pricing rules without engineering support.
- **CRUD Workflow Engineering:** Engineered React-based CRUD workflows connected to RDS/PostgreSQL via REST APIs, handling create/update validation, optimistic UI updates, and error rollback for a smoother editing experience.
- **UI/UX Development:** Led front-end UI/UX improvements across core product screens, translating design specs into polished, reusable React components used across multiple client-facing configurators.

Software Engineer | May 2021 – Feb 2023

- **Serverless API Development:** Built AWS Lambda APIs in Python with strong data integrity and scalability, supporting growing application workloads.
- **Frontend Engineering:** Developed React-based interfaces integrated with cloud services, improving responsiveness and user experience.
- **Cloud Asset Delivery:** Optimized 3D asset delivery via S3 and CloudFront, reducing asset load latency by ~25%.
- **Backend Automation:** Developed Python automation scripts for internal workflows, reducing manual operational overhead.
- **Testing Foundations:** Contributed to early testing infrastructure, improving code quality and deployment confidence.

Frontend Developer - Sanje Solutions | June 2020 – October 2020

- **Frontend Development:** Built responsive interfaces using React, HTML, CSS, and Bootstrap.
- **UI Implementation:** Translated design mockups into reusable, production-ready components with cross-browser compatibility.
- **Performance & UX:** Improved UI consistency and performance across client-facing applications.

Research Engineer - NUST (AI Systems & Web Systems) | June 2019 – October 2019

- **AI + Web Integration:** Developed web interfaces to visualize model outputs and enable interaction with AI systems.
- **Research Collaboration:** Worked with academic teams to translate research problems into technical implementations.
- **ML Pipelines Exploration:** Explored integration of machine learning workflows with lightweight web-based systems.

PROJECTS

3D Truck Configurator

- Built an interactive 3D product configurator using React, Three.js, and React Three Fiber, enabling real-time customization of vehicle color, parts, and environment lighting via PBR materials and HDRI maps.

- Architected a cross-window postMessage API enabling seamless third-party iframe embedding, exposing commands for live color/part updates, full config load, and state reset.
- Implemented state management with Zustand and deployed to production via AWS S3 + CloudFront.

Leaf Lab

- Developed an interactive educational web app teaching kids photosynthesis through a hands-on simulation (add CO₂/water, charge sunlight, run the reaction, see glucose/oxygen output).
- Built with React, TypeScript, Vite, Tailwind CSS, and Framer Motion; set up automated GitHub Actions deployment pipeline to GitHub Pages.

Generative 3D Environment Reconstruction Platform

- Developed workflows for reconstructing immersive 3D scenes from images using AI-generated assets, depth estimation, procedural scene assembly, automated camera placement, and Unreal Engine rendering.
- Integrated Blender, Unreal Engine, Claude Code, Florence-2, and modern image-to-3D models to accelerate virtual production workflows.

Pro Studio - 3D Room Layout Editor

- Built a browser-based 3D room design editor with first-person walkthrough, orbit, and top-down navigation modes, running entirely client-side with no backend
- Implemented an interactive scene editor with transform gizmos (move/rotate/scale), collision detection, snap-to-surface placement, and full undo/redo history
- Developed a live material editor and day/night lighting system with independent light-fixture control
- Built shareable layout links by encoding full scene and lighting state into URLs, plus PNG screenshot export and local-storage persistence
- Parsed GLB 3D assets at runtime to auto-generate a furniture catalog, distinguishing fixed architecture from placeable objects by node naming convention

ChatLLM.ai RAG Benchmark Suite

- Built an end-to-end benchmark suite for an Islamic-knowledge RAG chatbot, covering retrieval accuracy, citation quality, multi-step reasoning, and hallucination resistance across 10 evaluation categories.
- Engineered data pipelines pulling structured data from external APIs and parsing unstructured PDF sources (700+ pages) into machine-readable, citation-traceable records.
- Developed a Python scoring harness with automated pass/fail/needs-review verdicts, integrating configurable API calls and keyword-based fact verification.

TECHNICAL SKILLS

- **AI & Machine Learning:** Python (Advanced), PyTorch, TensorFlow, OpenCV, Computer Vision, Deep Learning, Pandas, NumPy, Matplotlib, RAG Pipeline Auditing, Hallucination & Citation Testing, Automated Benchmark Design, LLM-as-System Evaluation Methodology
- **Web & Full-Stack:** JavaScript, TypeScript, React.js, Next.js, Node.js, Material UI, SWR, HTML5, CSS3.
- **Cloud & DevOps:** AWS (Lambda, API Gateway, S3, RDS, EC2, CloudFront), Docker, Git, GitHub, CI/CD, Unit Testing (Jest/Pytest).
- **3D Visualization:** Unreal Engine, Blender, Three.js, React Three Fiber, WebGL, Blender, 3DS MAX, GLTF Compression, GPU Acceleration, Level of Detail (LOD). Unreal Engine 5, Blender Python, Gaussian Splatting, USD Pipeline, MetaHuman, Procedural Environment Generation
- **Databases:** PostgreSQL (RDS), MySQL, SQL, Schema Design, Query Optimization, Data Validation.
- **Generative AI & Spatial Models:** Generative 3D Reconstruction (Meshy, Tripo 3D, Hunyuan3D), Diffusion Models, Image Segmentation Masks, Automated Camera & Object Placement Algorithms.